



SHORE

Examination Number:

Set:

2024 YEAR 12 MID-YEAR EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using non-erasable black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks: 70

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7–12)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Note: Any time you have remaining should be spent reviewing your answers.

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Section I

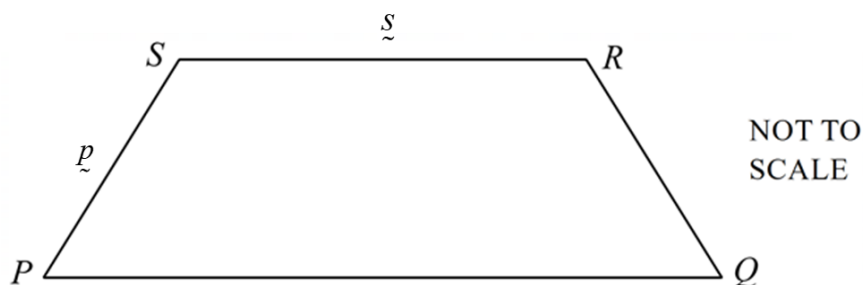
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 $PQRS$ is a trapezium where $\overrightarrow{PS} = \underline{p}$, $\overrightarrow{SR} = \underline{s}$, $\overrightarrow{PQ} = 2\overrightarrow{SR}$ and SR is parallel to PQ .



Which of the following is equivalent to $\overrightarrow{QS}$?

- A. $2\underline{s} + \underline{p}$
- B. $2\underline{s} - \underline{p}$
- C. $\underline{p} - 2\underline{s}$
- D. $-\underline{p} - 2\underline{s}$
- 2 Using the substitution $u = \tan x$, what is $\int \tan x \sec^2 x \, dx$?

- A. $\frac{1}{2} \tan x + C$
- B. $\frac{1}{2} \tan^2 x + C$
- C. $\tan x + C$
- D. $\tan^2 x + C$

- 3 Which of the following Cartesian equations represents the parametric equations $x = 2 \cos \theta - 1$ and $y = 2 + 2 \sin \theta$?

A. $\left(\frac{x-1}{2}\right)^2 + \left(\frac{y+2}{2}\right)^2 = 1$

B. $\left(\frac{x+1}{2}\right)^2 + \left(\frac{y-2}{2}\right)^2 = 4$

C. $(x+1)^2 + (y-2)^2 = 1$

D. $(x+1)^2 + (y-2)^2 = 4$

- 4 How many solutions for the equation $\sin 2x - \sin x = 0$ are in the domain $[0, 2\pi]$?

A. 2

B. 3

C. 4

D. 5

- 5 A spherical balloon is expanding so that its volume is increasing at a rate of 50 mm^3 per second. At what rate is the radius expanding when its radius is 10 mm ? Answer to 2 decimal places.

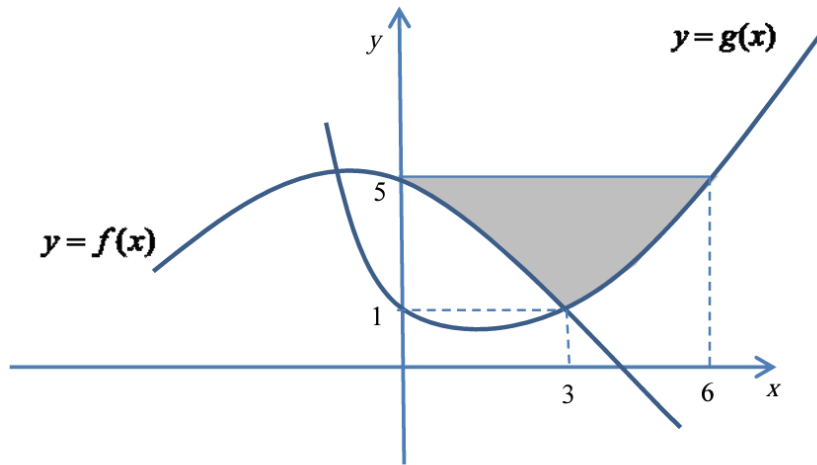
A. 0.02 mms^{-1}

B. 0.04 mms^{-1}

C. 0.08 mms^{-1}

D. 0.16 mms^{-1}

- 6 Consider the two functions $y = f(x)$ and $y = g(x)$ below.

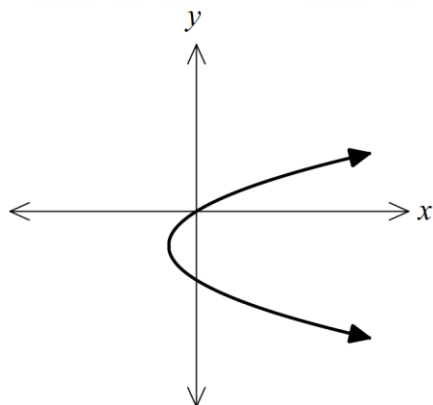


Which of the following definite integrals is equal to the area of the shaded region?

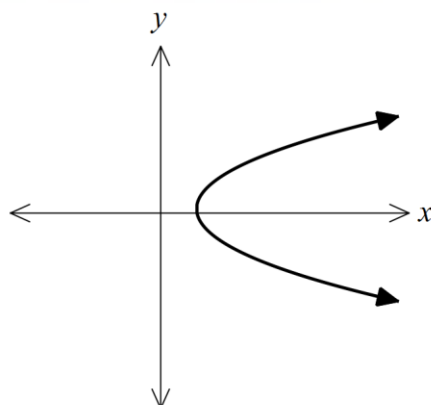
- A. $\int_1^5 (g(x) - f(x)) dx$
- B. $\int_1^5 (f(x) - g(x)) dx$
- C. $\int_0^3 (5 - g(x)) dx + \int_3^6 (5 - f(x)) dx$
- D. $\int_0^3 (5 - f(x)) dx + \int_3^6 (5 - g(x)) dx$
- 7 Which expression is $\int \frac{1}{u^2 + 6u + 10} du$ equivalent to?
- A. $\tan^{-1}(u + 3) + C$
- B. $\frac{1}{3} \tan^{-1}(u + 3) + C$
- C. $\tan^{-1}\left(\frac{u + 3}{\sqrt{10}}\right) + C$
- D. $\frac{1}{\sqrt{10}} \tan^{-1}(u + 3) + C$

- 8 Which of the following is a part of the graph given by parametric equations $x = t^2 + 2t$ and $y = t$?

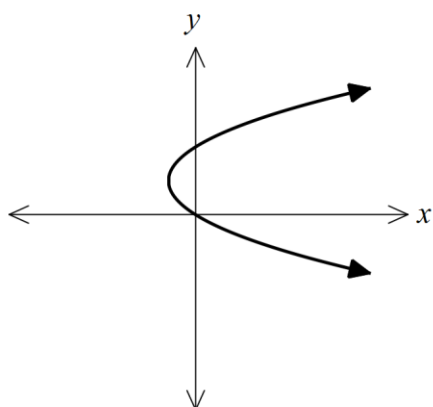
A.



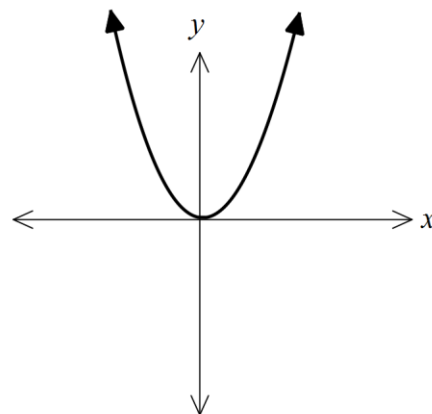
B.



C.



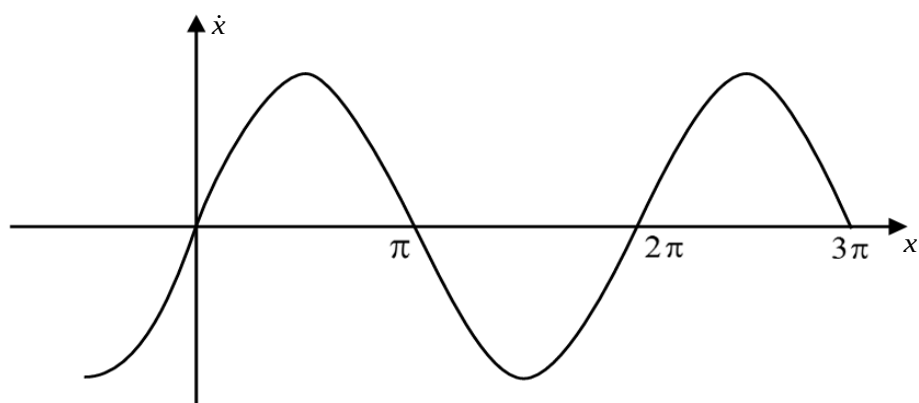
D.



- 9 What is the evaluation of $\int_{\frac{3}{4}}^{\frac{3}{4}} \cos^{-1} x \, dx$?

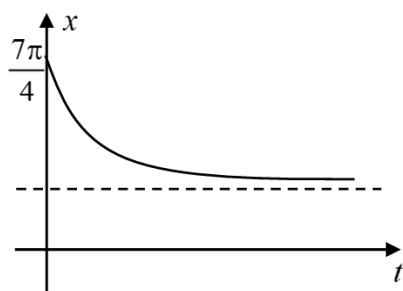
- A. $\frac{3}{2}$
 B. $\frac{3}{4}$
 C. $\frac{3\pi}{4}$
 D. $\frac{3\pi}{2}$

- 10 The velocity of a particle moving along the x axis is given by $\dot{x} = \sin x$ and is graphed below.

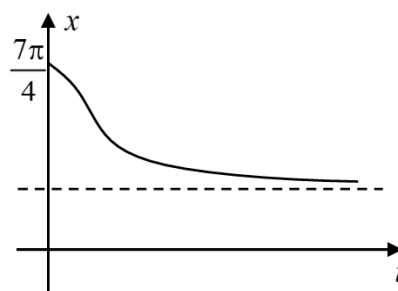


At $t = 0$, the particle is located at $x = \frac{7\pi}{4}$. Which of the following graphs best illustrates the motion of the particle for $t \geq 0$? (Graphs not drawn to scale.)

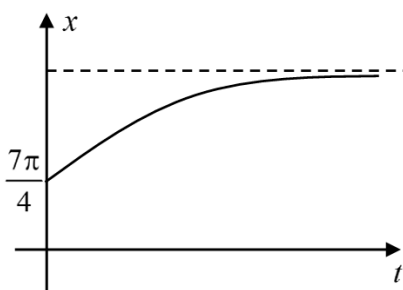
A.



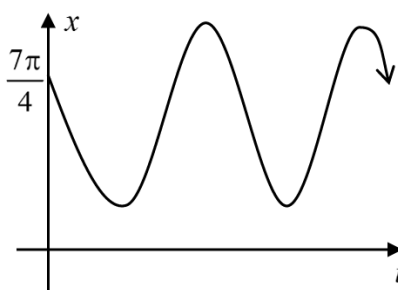
B.



C.



D.



End of Section 1

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

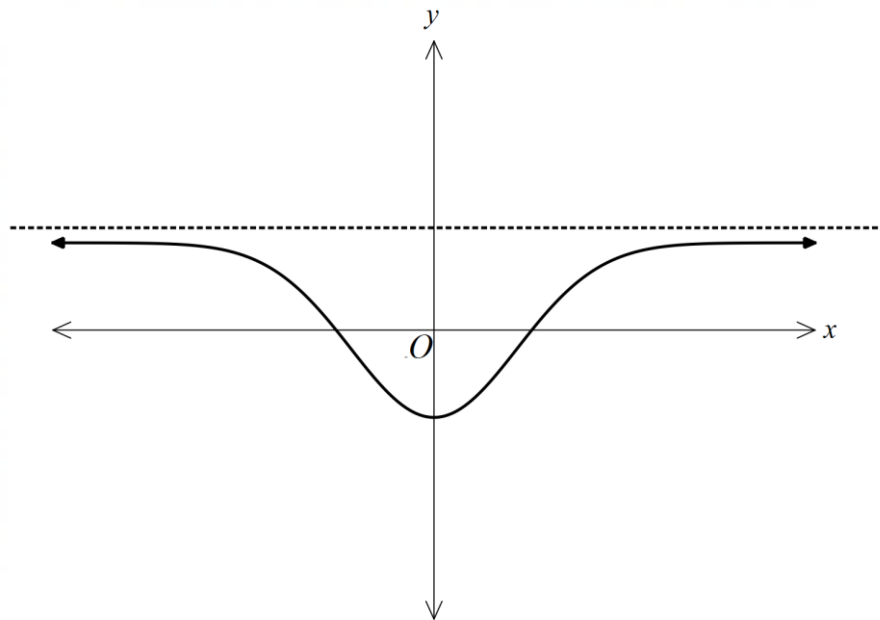
For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Use a SEPARATE writing booklet.

- (a) Given $\underline{u} = (-10, -1)$, find the unit vector in the same direction of $\underline{u}$. 1
- (b) Solve the inequality $4 \leq \frac{8}{x-3}$. 3
- (c) Given $\cos A = \frac{7}{9}$ and $\sin B = \frac{1}{3}$, where A and B are acute, using an appropriate trigonometric identity, show that $\cos 2B = \cos A$. 2
- (d) Consider the function $f(x) = \frac{x}{x+2}$.
- (i) Show that $f'(x) > 0$ for all x in the domain and hence explain why the function has an inverse function. 2
- (ii) Find the equation for $f^{-1}(x)$. 2
- (e) Use the substitution $u = \log_e x$ to find $\int_e^{e^2} \frac{1}{x(\log_e x)^2} dx$. 3

For question (f), answer parts (i) and (ii) on the separate answer sheet provided. Submit the answer sheet inside the booklet for question 11.

- (f) The diagram below shows the graph of $y = f(x)$ where $f(x) = 1 - 2e^{-x^2}$.



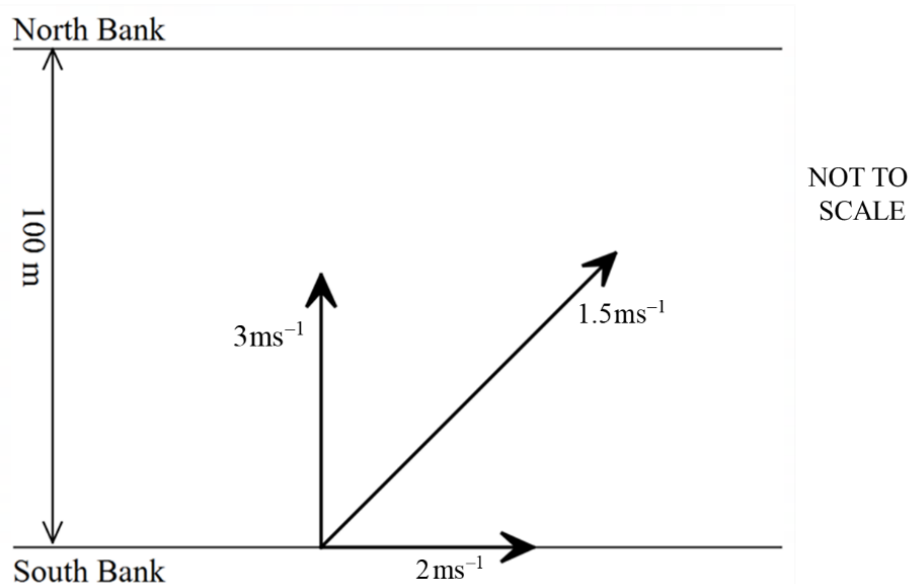
- (i) Find the equation of the horizontal asymptote. **1**
- (ii) The graph of this function **is also on a separate answer sheet**. On the same set of axes **on the answer sheet**, sketch $y = \sqrt{f(x)}$ showing the asymptote and the co-ordinates of any points of intersection between $y = f(x)$ and $y = \sqrt{f(x)}$. **2**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) Simon intends to row a boat from the south bank of a river to meet his friend on the north bank. The river is 100 metres wide. Simon's rowing speed is 3ms^{-1} . The river is flowing east at a rate of 2ms^{-1} .

The boat is also being impacted by a blowing wind from the south-west, which pushes it at 1.5ms^{-1} . Simon starts rowing across the river by steering the boat such that it is perpendicular to the south bank.



- (i) Show that the resultant velocity $\underline{v}$, of Simon's boat can be expressed as the component vector 2

$$\underline{v} = (2 + \sqrt{1.125})\underline{i} + (3 + \sqrt{1.125})\underline{j}$$

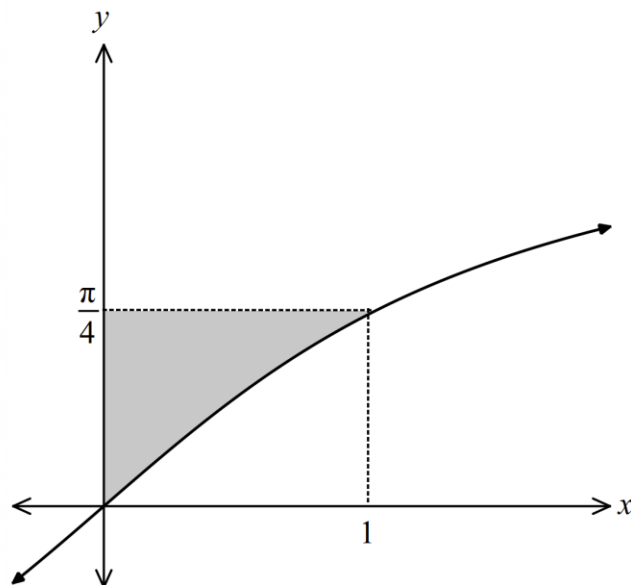
- (ii) Calculate the speed of the boat, correct to 1 decimal place. 1
- (iii) Determine the distance travelled from Simon's starting point to his landing point and how long it will take him to reach the north bank. 3

- (b) Assuming that people have, at most, 200,000 hairs on their head, prove that in a city of 8 million people, there are at least two people with exactly the same number of hairs on their head. 2

- (c) Given that the roots of $x^2 - 2x - 1 = 0$ are $\tan \alpha$ and $\tan \beta$, determine the value of $\alpha + \beta$. 2

(d) (i) Show that $\frac{d}{dy}(\ln(\cos y)) = -\tan y$. **1**

(ii) The diagram below shows a portion of the graph of $y = \tan^{-1} x$. Using **4**
part (i) show that its shaded area is given by $A = \frac{1}{2} \ln 2$ units².



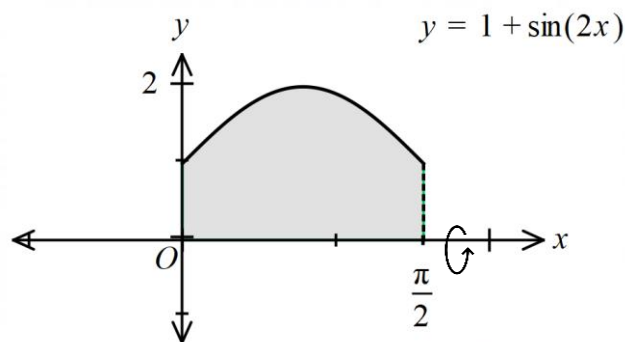
End of Question 12

Question 13 (14 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $\sin(M - 2N)\cos(M + 2N) = \frac{1}{2}(\sin 2M - \sin 4N)$. 2

(ii) Hence show that the exact value of $\sin(22.5^\circ)\cos(7.5^\circ) = \frac{2 + \sqrt{6} - \sqrt{2}}{8}$. 4

(b) The graph of $y = 1 + \sin(2x)$ is shown below for $0 \leq x \leq \frac{\pi}{2}$.



The shaded region is rotated about the x axis to form a solid of revolution. Find the volume of the solid leaving your answer in exact form. 4

(c) Consider the points $P(a, 2a)$, $Q(-a, 5a)$, $R(3a, 4a)$ and $S(9a, 12a)$ where a is a positive real number.

(i) Express $\overrightarrow{PQ}$ in component form. 1

(ii) Given the length of the projection of $\overrightarrow{PQ}$ onto $\overrightarrow{RS}$ is 12, find the value of a . 3

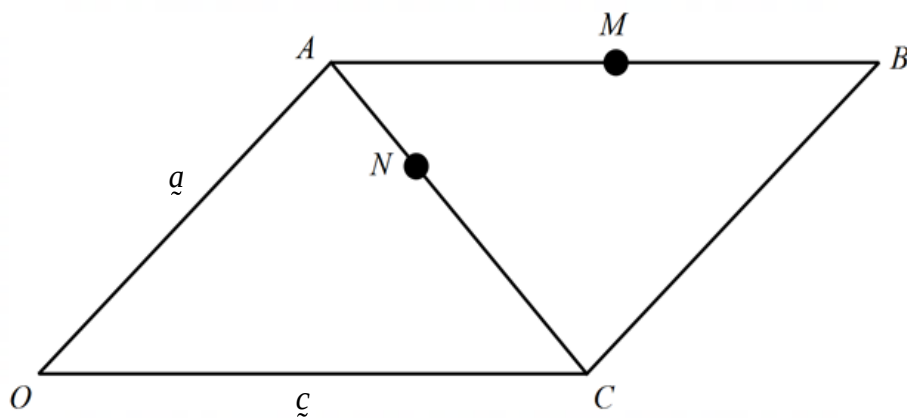
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) In the expansion of $(1+x)^n$, the coefficient of x^4 is twice the coefficient of x^3 . Find the value of n . 3
- (b) Find the derivative of $y = \cos^{-1}(\sqrt{2t-1})$. Leave answer in simplified form. 2
- (c) Use the principle of mathematical induction to prove that $2^{n+1}(5+7^n)$ is divisible by 6 for all positive integers n . 3
- (d) Let each different arrangement of all the letters in PERMUTATION be called a word.
- (i) Find an expression for the number of words possible. 1
- (ii) If a word is selected at random, find the probability that all the vowels are together. 2
- (iii) How many words are there in which the vowels occur in the order AEIOU from left to right though not necessarily together? 1
- (e) In the figure, $OABC$ is a parallelogram where the point M is the midpoint of AB . 3

The point N lies on the diagonal AC so that $AN : NC = 1 : 2$.

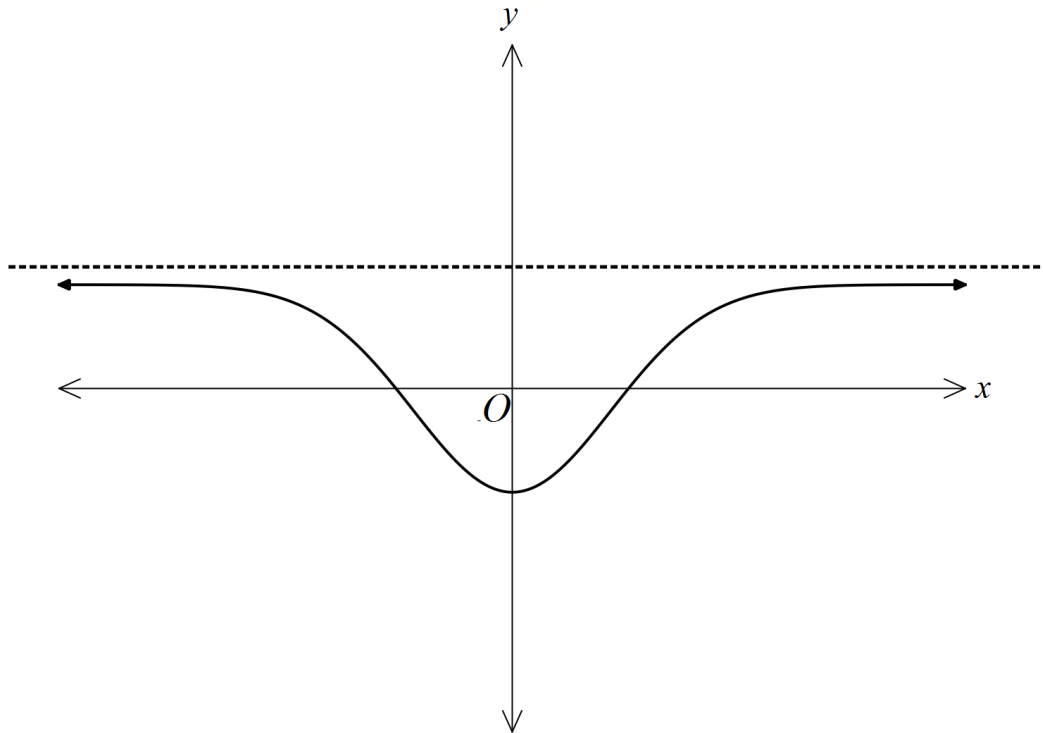
Let $\overrightarrow{OA} = \underline{a}$, and $\overrightarrow{OC} = \underline{c}$. Show that O , N and M are collinear.



☺ END OF PAPER ☹

Insert this sheet inside your answer booklet for Question 11

- f) The diagram below shows the graph of $y = f(x)$ where $f(x) = 1 - 2e^{-x^2}$.



- (i) Find the equation of the horizontal asymptote.

1

.....

.....

.....

.....

- (ii) On the same set of axes above, sketch $y = \sqrt{f(x)}$ showing the asymptote and the co-ordinates of any points of intersection between $y = f(x)$ and $y = \sqrt{f(x)}$.

2



SHORE

2024

Year 12 Mid-Year Examination

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Set

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Examination Number

Mathematics Extension 1

Section I Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ A. 2 B. 6 C. 8 D. 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

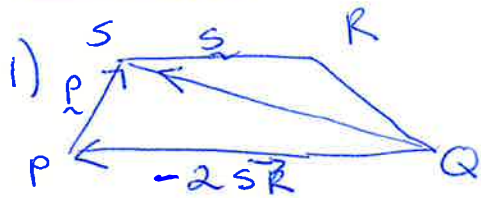
A ☒ B ☒ C ☐ D ☐
 correct

Start → Here

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Year 12 Ext 1 Mid Year Exam 2024

Multiple Choice:



$$\begin{aligned} & -2s\vec{R} + p \\ & = p - 2s\vec{R} \\ & = p - 2\vec{s} \end{aligned}$$

(C)

2) $\int \tan(x) \sec^2 x dx$

$$u = \tan x$$

$$\frac{du}{dx} = \sec^2 x$$

$$du = \sec^2 x dx$$

$$\int u du$$

$$= \frac{u^2}{2} + C$$

$$= \frac{1}{2} \tan^2 x + C$$

(B)

3) $x = 2\cos\theta - 1$ — (1)

$y = 2 + 2\sin\theta$ — (2)

from (1) $\cos\theta = \frac{x+1}{2}$

(2) $\sin\theta = \frac{y-2}{2}$

(D)

using $\sin^2\theta + \cos^2\theta = 1$

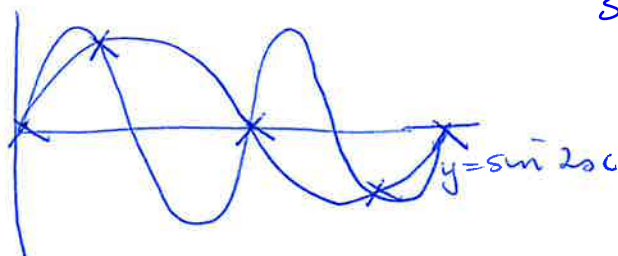
$$\left(\frac{x+1}{2}\right)^2 + \left(\frac{y-2}{2}\right)^2 = 1$$

$$(x+1)^2 + (y-2)^2 = 4$$

4) $\sin 2x - \sin x = 0$

OR

$$\begin{aligned} 2\sin x \cos x - \sin x &= 0 \\ \sin x (2\cos x - 1) &= 0 \end{aligned}$$



graphs intersect at 5 points

(D)

$$\sin x = 0 \text{ or } \cos x = \frac{1}{2}$$

3 solns

2 solns

$\therefore 5 \text{ solns}$

MCs:

C
B
D
D
B
D
A
A
C
B

$$5) \quad \frac{dV}{dt} = 50 \quad V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dr}{dV} = \frac{1}{4\pi r^2}$$

$$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{4\pi r^2} \times 50$$

$$= \frac{50}{4\pi r^2}$$

when $r=10$ $\frac{dr}{dt} = ?$

$$\frac{dr}{dt} = \frac{50}{4 \cdot \pi \cdot 10^2}$$

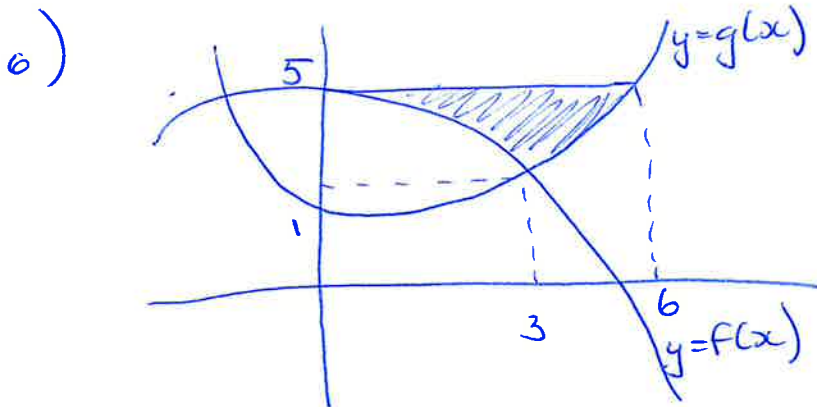
$$= \frac{50}{400\pi}$$

$$= \frac{1}{8\pi}$$

$$= 0.0397 \dots$$

$$= 0.04 \text{ mm/s (2dp)}$$

(B)



$$\int_0^3 (5 - f(x)) dx + \int_3^6 (5 - g(x)) dx$$

(D)

$$7) \int \frac{1}{u^2 + 6u + 10} du$$

$$u^2 + 6u + 10 = u^2 + 6u + 3^2 - 3^2 + 10 \quad (\text{CTS})$$

$$= (u+3)^2 + 1$$

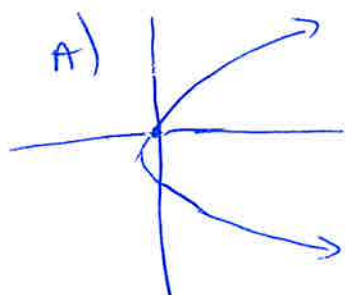
$$\text{Hence } \int \frac{1}{u^2 + 6u + 10} du$$

$$= \int \frac{1}{(u+3)^2 + 1} du$$

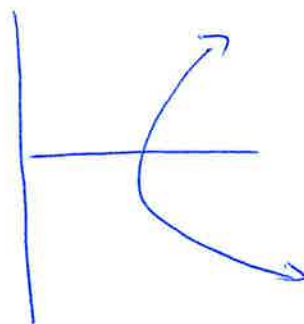
$$= \tan^{-1}(u+3) + C$$

(A)

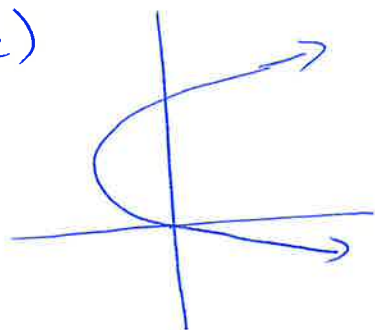
8)



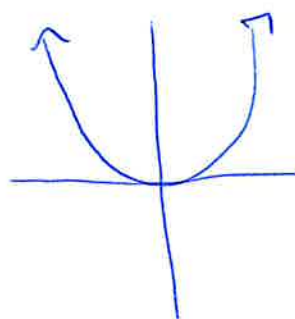
B)



C)



D)

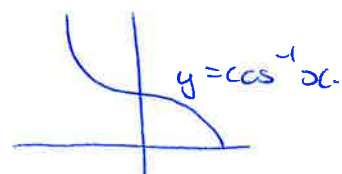
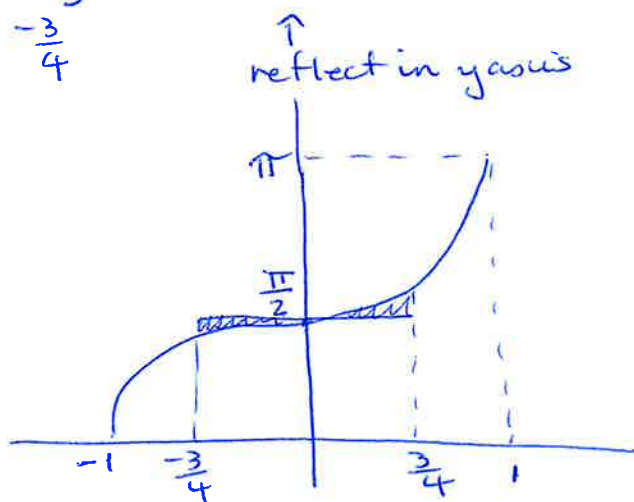


(A)

$x(t) = t^2 + 2t$ and $y(t) = t$
 when $t=0$ $x=0$ $y=0$ $(0,0)$ NOT B
 $t=1$ $x=3$ $y=1$ $(3,1)$ hard to tell
 $t=-1$ $x=-1$ $y=-1$ $(-1,-1)$

only $\uparrow$ A goes through

9) $\int_{-\frac{3}{4}}^{\frac{3}{4}} \cos^{-1}(6x) dx$



shading equal in size
given symmetry of
graph.

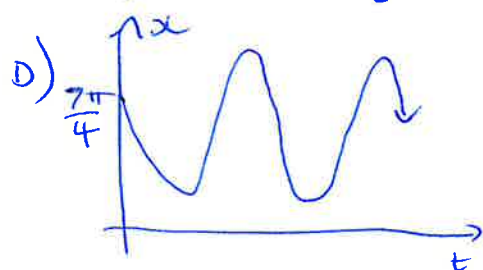
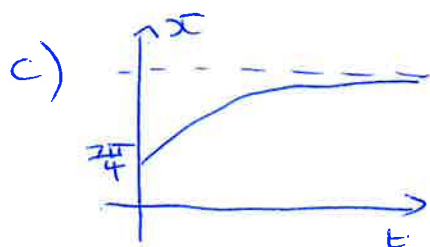
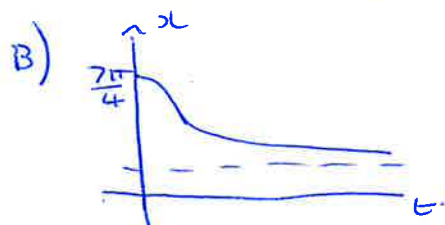
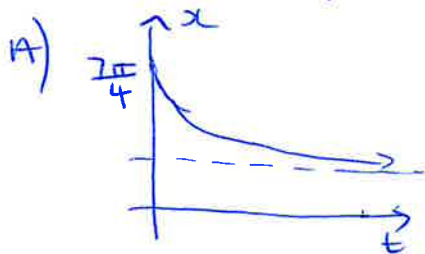
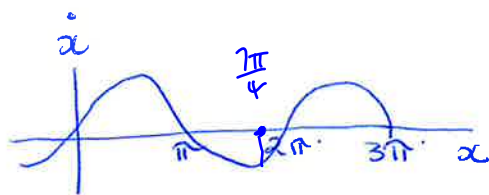
$$\int_{-\frac{3}{4}}^{\frac{3}{4}} \cos^{-1}(-x) dx = \text{Area rectangle}$$

$$= \frac{6}{4} \times \frac{\pi}{2}$$

$$= \frac{3\pi}{4}$$

(C)

10)



The graph shows that at $x = \frac{7\pi}{4}$
the velocity is negative so the
particle moves to the left (x decreases).
As x decreases at first the velocity
becomes more negative (the particle
moves at a faster rate towards the
left) then less negative (the particle
slows down). As the particle
moves towards $x = \pi$ the particle
moves at an ever slower rate
towards the left.

(B)

11a) Given $\underline{u} = (-10, -1)$ find the unit vector in the same direction of $\underline{u}$.

$$\hat{\underline{u}} = \frac{\underline{u}}{|\underline{u}|} \quad \underline{u} = -10\hat{i} - \hat{j}$$

$$\begin{aligned}\hat{\underline{u}} &= \frac{-10\hat{i} - \hat{j}}{\sqrt{10^2 + 1^2}} \\ &= \frac{-10\hat{i} - \hat{j}}{\sqrt{101}}\end{aligned}$$

$$\hat{\underline{u}} = \frac{-10}{\sqrt{101}} \hat{i} - \frac{1}{\sqrt{101}} \hat{j}$$

OR

$$\frac{-10\sqrt{101}}{101} \hat{i} - \frac{\sqrt{101}}{101} \hat{j}$$

OR

$$\left(\frac{10\sqrt{101}}{101}, \frac{\sqrt{101}}{101} \right)$$

OR

$$\frac{1}{\sqrt{101}} \begin{pmatrix} -10 \\ -1 \end{pmatrix}$$

$$11b) \quad 4^{x(x-3)} \leq \frac{8}{x-3} \quad x \neq 3$$

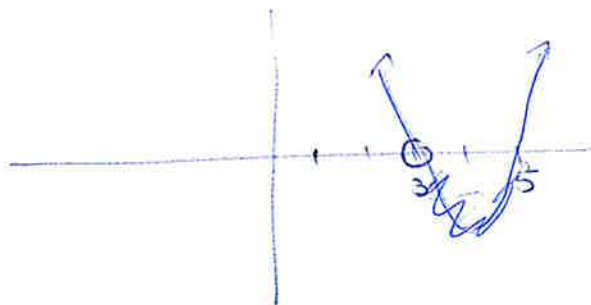
$$4(x-3)^2 \leq 8(x-3)$$

$$4(x-3)^2 - 8(x-3) \leq 0$$

$$4(x-3)(x-3-2) \leq 0$$

$$4(x-3)(x-5) \leq 0$$

$$x \neq 3 \text{ or } x = 5$$



$$3 < x \leq 5$$

OR

$$[3, 5]$$

OR

$$4 \leq \frac{8}{x-3}$$

Critical points



$$x-3 \neq 0 \\ x \neq 3$$

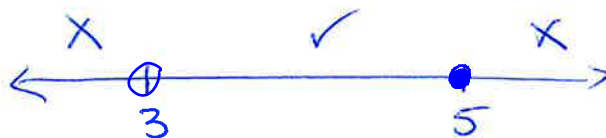


$$4x-12 = 8$$

$$4x = 20$$

$$x = 5$$

Test



$$x = 0$$

$$4 \leq \frac{8}{0-3} \times$$

$$x = 4$$

$$4 \leq \frac{8}{4-3} \checkmark$$

$$x = 6$$

$$4 \leq \frac{8}{6-3} \times$$

$$\therefore 3 < x \leq 5$$

11c) $\cos A = \frac{7}{9}$ $\sin B = \frac{1}{3}$

$$\begin{aligned}\text{LHS} &= \cos 2B \\ &= 1 - 2\sin^2 B \\ &= 1 - 2\left(\frac{1}{3}\right)^2 \\ &= 1 - \frac{2}{9} \\ &= \frac{7}{9}\end{aligned}$$

$$\begin{aligned}&= \cos A \\ &= \text{RHS}\end{aligned}$$

$$\therefore \cos 2B = \cos A$$

A, B acute, show
 $\cos 2B = \cos A$

$$\left[\begin{aligned}\cos 2B &= \cos^2 B - \sin^2 B \\ &= 1 - \sin^2 B - \sin^2 B \\ &= 1 - 2\sin^2 B\end{aligned} \right]$$

11d)

$$f(x) = \frac{x}{x+2} \quad x \neq -2$$

$$i) \quad f'(x) = \frac{(x+2) \cdot 1 - 1 \cdot x}{(x+2)^2}$$

$$= \frac{2}{(x+2)^2} > 0 \text{ for all values of } x.$$

As $(x+2)^2 \geq 0$ then $f'(x) > 0$ for all values of x .

$\therefore f(x)$ is monotonic increasing and hence has

an inverse function.

OR

$$ii) \quad y = \frac{x}{x+2}$$

$$x = \frac{y}{y+2}$$

$$xy + 2x = y$$

$$xy - y = -2x$$

$$y(x-1) = -2x$$

$$y = \frac{-2x}{x-1}$$

$$\therefore f^{-1}(x) = \frac{-2x}{x-1}$$

As $f(x)$ always has an increasing gradient, it passes the horizontal line test and hence has an inverse function.

$$11e) \int_e^{e^2} \frac{1}{x(\log_e x)^2} dx \quad \text{let } u = \log_e x$$

$$u = \ln x$$

$$\text{when } x = e \quad u = \ln e = 1$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\text{when } x = e^2 \quad u = \ln e^2 = 2 \ln e = 2$$

$$du = \frac{dx}{x}$$

$$\text{Hence } \int_e^{e^2} \frac{1}{x(\log_e x)^2} dx$$

← All in 'x'

$$= \int_1^2 \frac{du}{u^2}$$

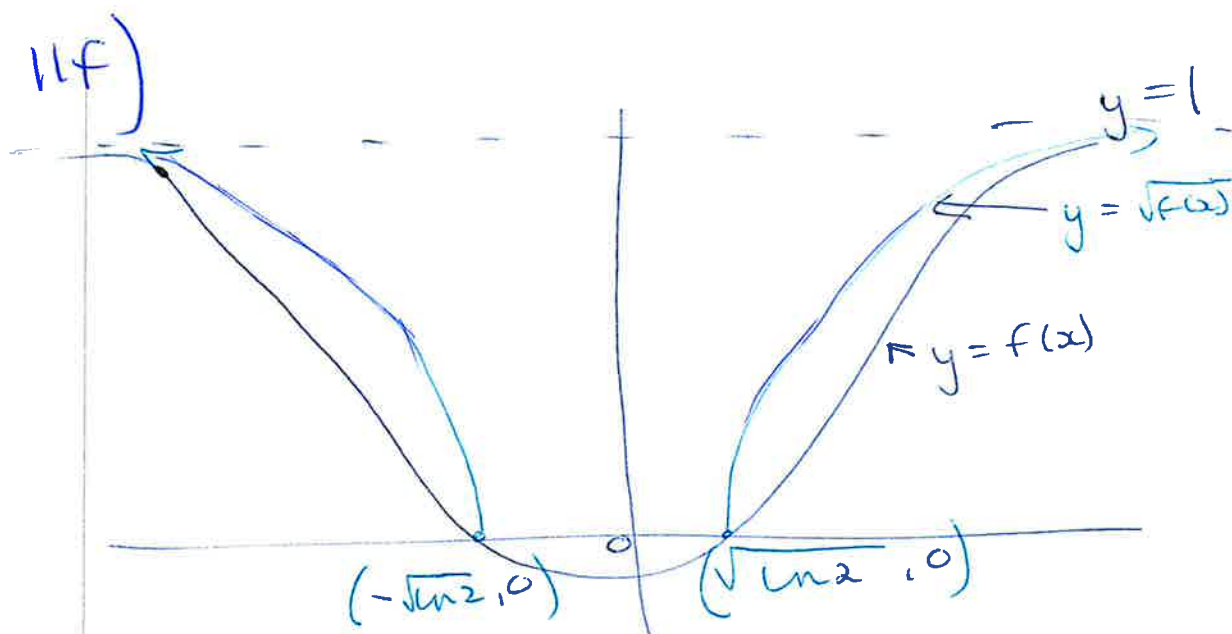
← All in 'u'.

$$= \left[-u^{-1} \right]_1^2$$

$$= -2^{-1} + 1^{-1}$$

$$= -\frac{1}{2} + 1$$

$$= \frac{1}{2}$$



- i) $f(x) = 1 - 2e^{-x^2}$
 $x \rightarrow \infty$ from above and below $e^{x^2} \rightarrow \infty$
 $\therefore \frac{2}{e^{x^2}} \rightarrow 0$
 $\therefore f(x) \rightarrow 1$
 $\therefore$ horizontal asymptote is $y = 1$

ii)

$$0 = 1 - 2e^{-x^2}$$

$$\frac{1}{2} = \frac{1}{e^{x^2}}$$

$$e^{x^2} = 2$$

$$x^2 = \ln 2$$

$$x = \pm \sqrt{\ln 2}$$

$$\frac{1}{2} = e^{-x^2}$$

$$\ln\left(\frac{1}{2}\right) = \ln(e^{-x^2})$$

$$\therefore \ln\left(\frac{1}{2}\right) = -x^2$$

$$-\ln\left(\frac{1}{2}\right) = x^2$$

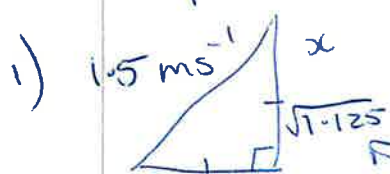
Common answer: $x = \pm \sqrt{-\ln\left(\frac{1}{2}\right)}$

Show that $-\ln\left(\frac{1}{2}\right) > 0$. $\downarrow$
 $-\ln(2^{-1}) = \ln 2$

$$\therefore x = \pm \sqrt{\ln 2}$$

12a)

Components of wind vector.



$$(1.5)^2 = 2x^2$$

$$x^2 = 1.125$$

$$x = \pm \sqrt{1.125}$$

$$x = \sqrt{1.125} \text{ only since } x \geq 0$$

Wind vector is $\sqrt{1.125} \hat{i} + \sqrt{1.125} \hat{j}$

Rowing vector is $0 \hat{i} + 3 \hat{j}$

Water flow vector is $2 \hat{i} + 0 \hat{j}$

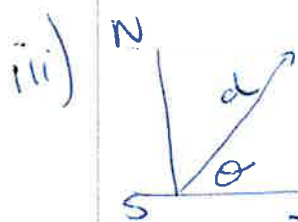
$$\begin{aligned} \text{Resultant velocity} &= 2 \hat{i} + 0 \hat{j} + 0 \hat{i} + 3 \hat{j} + \sqrt{1.125} \hat{i} + \sqrt{1.125} \hat{j} \\ &= (2 + \sqrt{1.125}) \hat{i} + (3 + \sqrt{1.125}) \hat{j} \\ &\text{as required.} \end{aligned}$$

ii) Velocity is equal to magnitude of resultant vector in i)

$$v = \sqrt{(2 + \sqrt{1.125})^2 + (3 + \sqrt{1.125})^2}$$

$$v = 5.0849387 \dots$$

$$v = 5.1 \text{ ms}^{-1} \text{ (1.d.p.)}$$

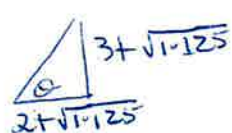


$$\tan \theta = \frac{3 + \sqrt{1.125}}{2 + \sqrt{1.125}}$$

$$\theta = \tan^{-1} \left(\frac{3 + \sqrt{1.125}}{2 + \sqrt{1.125}} \right)$$

$$= 52.993401 \dots$$

$$\approx 53^\circ$$



'd' is distance from start to finish.

$$\sin \theta = \frac{100}{d}$$

$$d = \frac{100}{\sin 53}$$

$$d = 125.21 \dots$$

$$d \approx 125 \text{ m}$$

$\therefore$ distance from starting to landing point is 125 m.



$$\text{Time taken} = \frac{\text{distance}}{\text{speed}}$$

$$= \frac{125.21 \dots}{5.084 \dots}$$

$$= 24.6 \text{ seconds}$$

$\therefore$ time taken to reach North bank
is 24.6 seconds

OR.

$$\text{Time taken, } t_f = \frac{\text{vertical distance}}{\text{vertical component of velocity}}$$

$$= \frac{100}{3 + \sqrt{1.125}}$$

$$= 24.6265 \dots$$

$$= 24.63$$

$$\text{Distance} = |v| \times t_f$$

$$= 24.63 \times 5.1$$

$$= 125.61$$

$$= 125.6 \text{ m.}$$

12b)

Assume that at most, people have 200000 hairs on their head. Prove that in a city of 8 million people, there are at least ¹ two people with exactly the same number of hairs on their head.

pigeons = 8 million people

pigeons holes = 0, 1 --- 200 000 hairs

8 million people to be sorted into 200000+1 pigeon holes.

Since $\frac{8000000}{200001} = 39.998$, by pigeon

hole principle there are 40 people with the same number of hairs on their head.

Hence there are at least 2 people (in fact there's closer to 40) with the same number of hairs on their head.

12c)

$x^2 - 2x - 1 = 0$ $\tan \alpha$ and $\tan \beta$ are roots.

sum roots: $\tan \alpha + \tan \beta = 2$

product roots: $\tan \alpha \tan \beta = -1$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan(\alpha + \beta) = \frac{2}{1+1}$$

$$\tan(\alpha + \beta) = 1$$

$$\alpha + \beta = \frac{\pi}{4}$$

$$\therefore \alpha + \beta = \frac{\pi}{4}$$

OR.

$$x = \frac{2 \pm \sqrt{4+4}}{2}$$

$$= 1 \pm \sqrt{2}$$

$$\tan \alpha = 1 + \sqrt{2} \quad \tan \beta = 1 - \sqrt{2}$$

$$\alpha = \tan^{-1}(1 + \sqrt{2}) \quad \beta = \tan^{-1}(1 - \sqrt{2})$$

$$= 67.5$$

$$= -22.5$$

$$\alpha + \beta = 67.5 - 22.5$$

$$= 45$$

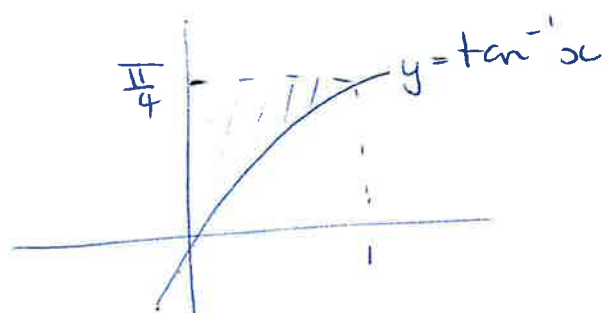
$$= \frac{\pi}{4}$$

12d)

i) Show $\frac{d}{dy}(\ln(\cos y)) = -\tan y$

$$\begin{aligned} \text{LHS} &= \frac{d}{dy}(\ln(\cos y)) \\ &= \frac{1}{\cos y} \cdot -\sin y \\ &= \frac{-\sin y}{\cos y} \\ &= -\tan y \\ &= \text{RHS} \end{aligned}$$

ii)



$$y = \tan^{-1} x$$

$$x = \tan y$$

$$A = \int_0^{\pi/4} \tan y$$

$$= - \int_0^{\pi/4} -\tan y \quad (\text{using (i)})$$

$$= - \left[\ln(\cos y) \right]_0^{\pi/4}$$

$$= - \left(\ln \cos \frac{\pi}{4} - \ln(\cos 0) \right)$$

$$= - \left(\ln \frac{1}{\sqrt{2}} - \ln 1 \right)$$

$$= - \ln 2^{-1/2}$$

$$= \frac{1}{2} \ln 2 \text{ units}^2 \text{ as required}$$

13a) Show $\sin(M-2N)\cos(M+2N) = \frac{1}{2}(\sin 2M - \sin 4N)$

$$\text{LHS} = \sin(M-2N)\cos(M+2N)$$

$$= \frac{1}{2} \left(\sin(M-2N+M+2N) + \sin(M-2N-M-2N) \right)$$

$$= \frac{1}{2} (\sin 2M + \sin(-4N))$$

$$= \frac{1}{2} (\sin 2M - \sin 4N) \quad (\text{As sine is odd})$$

$$= \text{RHS as required.}$$

using
 $\sin A \cos B = \frac{1}{2}(\sin(A+B) + \sin(A-B))$

ii) $\sin(M-2N)\cos(M+2N) = \sin(22\frac{1}{2})\cos(7\frac{1}{2})$
 $M-2N = 22\frac{1}{2} \quad \text{--- (1)} \quad M+2N = 7\frac{1}{2} \quad \text{--- (2)}$

(1) + (2) $2M = 30$
 $M = 15 \Rightarrow 15 + 2N = 7\frac{1}{2}$
 $N = -\frac{15}{4}$

Hence $\sin 22\frac{1}{2}\cos 7\frac{1}{2} = \frac{1}{2}(\sin(2 \times 15) - \sin(4 \times -\frac{15}{4}))$
 $= \frac{1}{2}(\sin(30) - \sin(-15))$
 $= \frac{1}{2}(\sin 30 + \sin 15)$
 $= \frac{1}{2}(\sin 30 + \sin(45-30))$
 $= \frac{1}{2}(\sin 30 + \sin 45 \cos 30 - \sin 30 \cos 45)$
 $= \frac{1}{2}\left(\frac{1}{2} + \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{\sqrt{2}}\right)$
 $= \frac{\sqrt{2} + \sqrt{3} - 1}{4\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$
 $= \frac{2 + \sqrt{6} - \sqrt{2}}{4 \cdot 2}$
 $= \frac{2 + \sqrt{6} - \sqrt{2}}{8} \text{ as required}$

13b)

$$V = \pi \int_a^b y^2 dx \quad \text{from graph } a=0 \quad b=\frac{\pi}{2}$$

$$y = 1 + \sin 2x$$

$$y^2 = (1 + \sin 2x)^2$$

$$V = \pi \int_0^{\frac{\pi}{2}} (1 + \sin 2x)^2 dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (1 + \sin^2 2x + 2 \sin 2x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \left(1 + \frac{1}{2}(1 - \cos 4x) + 2 \sin 2x \right) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \left(\frac{3}{2} - \frac{1}{2} \cos 4x + 2 \sin 2x \right) dx$$

$$= \pi \left[\frac{3}{2} x - \frac{1}{8} \sin 4x - \cos 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\frac{3}{2} \times \frac{\pi}{2} - \frac{1}{8} \sin 2\pi - \cos 2\pi - 0 + 0 + \cos 0 \right]$$

$$= \pi \left(\frac{3\pi}{4} - 0 - -1 + 1 \right)$$

$$= \pi \left(\frac{3\pi}{4} + 2 \right)$$

$$= \frac{3\pi^2 + 8\pi}{4} \text{ units}^3$$

using
 $\sin^2 x = 1 - \cos^2 x$
 $\sin^2 2x = 1 - \cos^2 2x$
 $\sin^2 2x = \frac{1}{2}(1 - \cos 4x)$

13c)

$P(a, 2a)$, $Q(-a, 5a)$, $R(3a, 4a)$, $S(9a, 12a)$
 Length of projection is 12 ← scalar

$$i) \vec{PQ} = \begin{pmatrix} -a-a \\ 5a-2a \end{pmatrix} \therefore \vec{PQ} = -2a\hat{i} + 3a\hat{j} \\ = \begin{pmatrix} -2a \\ 3a \end{pmatrix}$$

ii) METHOD 1: scalar projection of $\vec{PQ}$ onto $\vec{RS}$

$$\text{Scalar}_{\text{proj}} = \frac{\vec{PQ} \cdot \vec{RS}}{|\vec{RS}|} \quad \vec{RS} = \begin{pmatrix} 9a-3a \\ 12a-4a \end{pmatrix} = \begin{pmatrix} 6a \\ 8a \end{pmatrix} = 6a\hat{i} + 8a\hat{j}$$

$$\text{Dot product } \vec{PQ} \cdot \vec{RS} = -2a \times 6a + 3a \times 8a \\ = -12a^2 + 24a^2 \\ = 12a^2$$

$$|\vec{RS}| = \sqrt{(6a)^2 + (8a)^2} \\ = \sqrt{36a^2 + 64a^2} \\ = \sqrt{100a^2} \\ = 10a \quad (\text{as } a > 0, \text{ given})$$

$$\therefore \frac{\vec{PQ} \cdot \vec{RS}}{|\vec{RS}|} = \frac{12a^2}{10a} \quad \text{however scalar projection of } \vec{PQ} \text{ onto } \vec{RS} \text{ is } 12 \text{ (given)}$$

$$12 = \frac{6a}{5}$$

$$a = 10$$

METHOD 2: using vector projection (not scalar)

$$\text{Vector}_{\text{proj}} = \frac{\vec{PQ} \cdot \vec{RS}}{|\vec{RS}|^2} \cdot \vec{RS}$$

$$= \frac{\begin{pmatrix} -2a \\ 3a \end{pmatrix} \begin{pmatrix} 6a \\ 8a \end{pmatrix} (6a\hat{i} + 8a\hat{j})}{(6a)^2 + (8a)^2} \quad \leftarrow \text{as per method 1}$$

$$= \frac{-12a^2 + 24a^2}{36a^2 + 64a^2} (6a\hat{i} + 8a\hat{j})$$

$$= \frac{12a^2}{100a^2} (6a\hat{i} + 8a\hat{j})$$

$$= \frac{3}{25} (6a\hat{i} + 8a\hat{j})$$

$$= \frac{18a\hat{i}}{25} + \frac{24a}{25} \hat{j}$$

Given projection length is 12

$$12 = \sqrt{\left(\frac{18a}{25}\right)^2 + \left(\frac{24a}{25}\right)^2}$$

$$12 = \sqrt{\frac{900a^2}{25^2}}$$

$$12 = \frac{30a}{25}$$

$$\frac{6a}{5} = 12$$

$$6a = 60$$

$$a = 10$$

14a)

$$(1+x)^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n x^n$$

$$\text{coeff } x^4 = 2 \times \text{coeff } x^3$$

$${}^nC_4 = 2 {}^nC_3$$

$$\frac{n!}{4!(n-4)!} = 2 \frac{n!}{3!(n-3)!}$$

$$\frac{3!(n-3)!}{4!(n-4)!} = 2 \frac{\cancel{n!}}{\cancel{n!}}$$

$$\frac{3!(n-3)(n-4)!}{4 \cdot 3!(n-4)!} = 2$$

$$\frac{n-3}{4} = 2$$

$$n-3 = 8$$

$$n = 11$$

14b)

$$y = \cos^{-1}(\sqrt{2t-1})$$

$$\frac{dy}{dx} = - \frac{f'(x)}{\sqrt{1-(f(x))^2}}$$

$$= \frac{-(2t-1)^{-\frac{1}{2}}}{\sqrt{1-(2t-1)}}$$

$$= - \frac{1}{(2t-1)^{\frac{1}{2}} \sqrt{2-2t}}$$

$$= - \frac{1}{(\sqrt{2t-1})(\sqrt{2-2t})}$$

$$f(x) = (2t-1)^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2} (2t-1)^{-\frac{1}{2}} \cdot 2$$

$$= (2t-1)^{-\frac{1}{2}}$$

$$(f(x))^2 = ((2t-1)^{\frac{1}{2}})^2$$

$$= 2t-1$$

14c)

Prove $2^{n+1}(5+7^n)$ is divisible by 6 for all n ,
positive integers

1) Prove statement true for $n=1$

$$\text{LHS} = 2^{1+1}(5+7^1)$$

$$= 2^2(12)$$

$$= 4 \times 2 \times 6 \text{ which is divisible by } 6$$

$\therefore$ true for $n=1$

2) Assume statement true for $n=k$

$$2^{k+1}(5+7^k) = 6M \quad \text{where } M \text{ \& } k \text{ are positive integers}$$

$$2^{k+1} \times 5 = 6M - 7^k(2^{k+1})$$

3) Prove statement true for $n=k+1$

$$2^{k+1+1}(5+7^{k+1}) = 6Q \quad \text{where } Q \text{ is a positive integer}$$

$$\text{LHS} = 2^{k+1+1}(5+7^{k+1})$$

$$= 2^{k+2}(5+7^{k+1})$$

$$= 2 \cdot 2^{k+1}(5+7^1 \cdot 7^k)$$

$$= 2 \cdot 5 \cdot 2^{k+1} + 2 \cdot 7 \cdot 2^{k+1} \cdot 7^k$$

$$= 2(6M - 7^k(2^{k+1})) + 14 \cdot 2^{k+1} \cdot 7^k \quad \text{by assumption.}$$

$$= 12M - 2 \cdot 7^k \cdot 2^{k+1} + 14 \cdot 2^{k+1} \cdot 7^k$$

$$= 12M + 12 \cdot 7^k \cdot 2^{k+1}$$

$$= 12(M + 7^k \cdot 2^{k+1})$$

$$= 6 \times 2(M + 7^k \cdot 2^{k+1})$$

$$= 6Q \quad \text{where } Q = 2(M + 7^k \cdot 2^{k+1})$$

$$= \text{RHS}$$

Statement is divisible by 6 $\therefore$ is true
for $n=k+1$

4) Therefore statement is true for all
positive integers n .

14 d)

i) PERMUTATION

$$\# \text{ words} = \frac{11!}{2!} = 19958400 \text{ words}$$

ii) P(all vowels together)

$$= \frac{\# \text{ ways vowels together}}{\text{total } \# \text{ ways}}$$

$$= \frac{\frac{7!}{2!} \times 5!}{\frac{11!}{2!}}$$

$$= \frac{7! \times 5!}{11!}$$

$$= \frac{1}{66}$$

all vowels
together E, U, A, I, O

PRMTTN

$\frac{7!}{2!}$ ways to
arrange group
(2 Ts)

$\times 5!$ ways to
arrange vowels

iii) _ _ _ _ _
11 positions

$\binom{11}{5}$ ways to choose 5 positions for vowels A, E, I, O, U
and only 1 way to arrange vowels.

$\frac{6!}{2!}$ ways to arrange consonants

$$\therefore \# \text{ words} = \binom{11}{5} \times \frac{6!}{2!} = 166320$$

OR

$\frac{1}{5!}$ of the arrangements will have vowels
in order $\therefore \# \text{ words} = \frac{1}{5!} \times \frac{11!}{2!} = 166320$

14e)

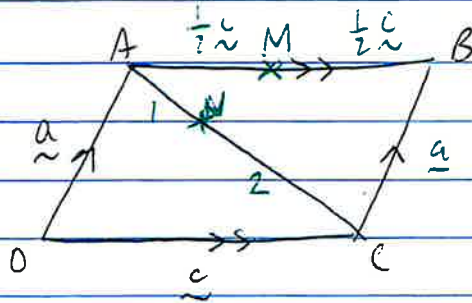
Vector Question

OABC is a parallelogram and the point M is the midpoint of AB.

The point N lies on the diagonal AC so that $AN:NC = 1:2$.

Let $\vec{OA} = \underline{a}$ and $\vec{OC} = \underline{c}$.

Deduce, showing reasoning, that O, N & M are collinear.



$$\begin{aligned}\vec{AC} &= \vec{AO} + \vec{OC} \\ &= -\underline{a} + \underline{c}\end{aligned}$$

$$\begin{aligned}\vec{AN} &= \frac{1}{3}\vec{AC} \\ &= \frac{1}{3}(-\underline{a} + \underline{c})\end{aligned}$$

$$\begin{aligned}\therefore \vec{ON} &= \vec{OA} + \vec{AN} \\ &= \underline{a} + \frac{1}{3}(-\underline{a} + \underline{c}) \\ &= \underline{a} - \frac{1}{3}\underline{a} + \frac{1}{3}\underline{c} \\ &= \frac{2}{3}\underline{a} + \frac{1}{3}\underline{c} \\ &= \frac{1}{3}[2\underline{a} + \underline{c}]\end{aligned}$$

$$\begin{aligned}\vec{NM} &= \vec{NA} + \vec{AM} \\ &= -\frac{1}{3}(-\underline{a} + \underline{c}) + \frac{1}{2}\underline{c} \\ &= \frac{1}{3}\underline{a} - \frac{1}{3}\underline{c} + \frac{1}{2}\underline{c} \\ &= \frac{1}{3}\underline{a} + \frac{1}{6}\underline{c} \\ &= \frac{1}{6}[2\underline{a} + \underline{c}]\end{aligned}$$

$\therefore \vec{ON} = \vec{NM}$, ~~and~~ are in the same direction, & share the point N.

$\therefore$ O, N & M are collinear.